GHAZI UNIVERSITY D.G KHAN

Assignment

Subject: Programming Fundamentals_Lab2

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Class: B.S C.S (1st Semester)

Roll number: 04

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LAB#2

TASK1:

Type and save the following program in visual studio. Run these program and observe their output.

```
1.#include<iostream>
using namespace std;
int main(){
        char letter;
letter = 'A';
cout<<letter<<"\n";
letter = 'A';
cout<<letter<<"\n";
cin>>letter;        return
0;

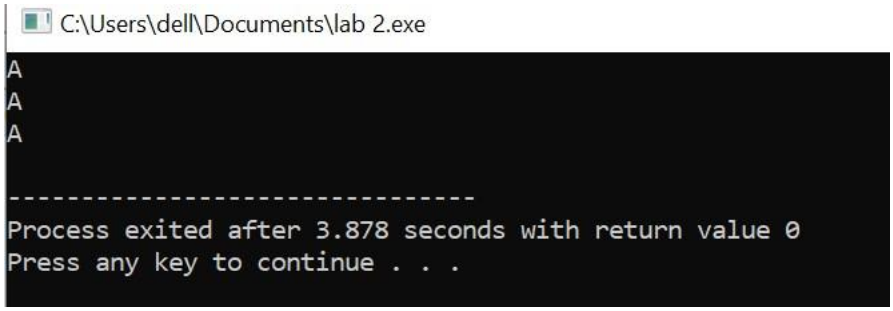
}
```

Output:

A

A

A



2.#include<iostream>

using namespace std;

int main(){

char letter;

letter = 65;

cout<<letter<<'\n';

letter = 66;

cout<<letter<<'\n';

cin>>letter; return

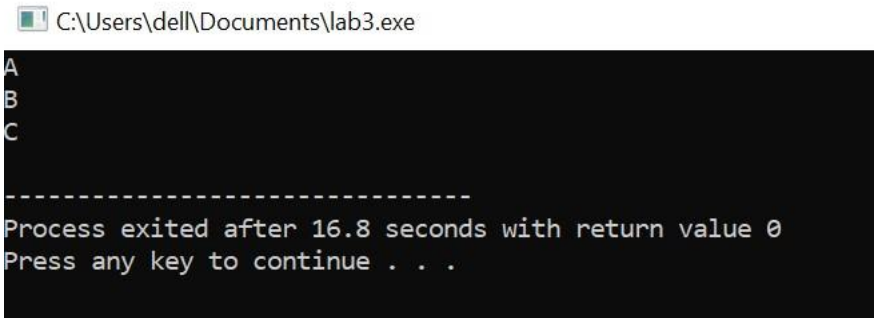
0;

}

OutPut:

A B

C



3. #include<iostream>

using namespace std;

int main(){

int letter; letter =

'A'; cout<<letter<<'\n';

letter = 'B';

cout<<letter<<'\n';

cin>>letter; return

0;

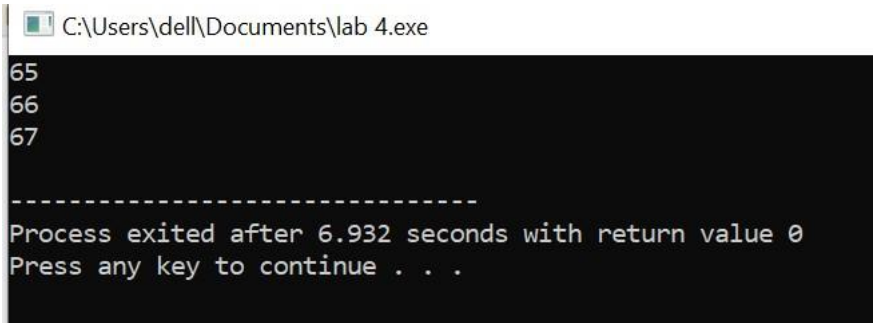
}

Output:

65

66

67



TASK 2:

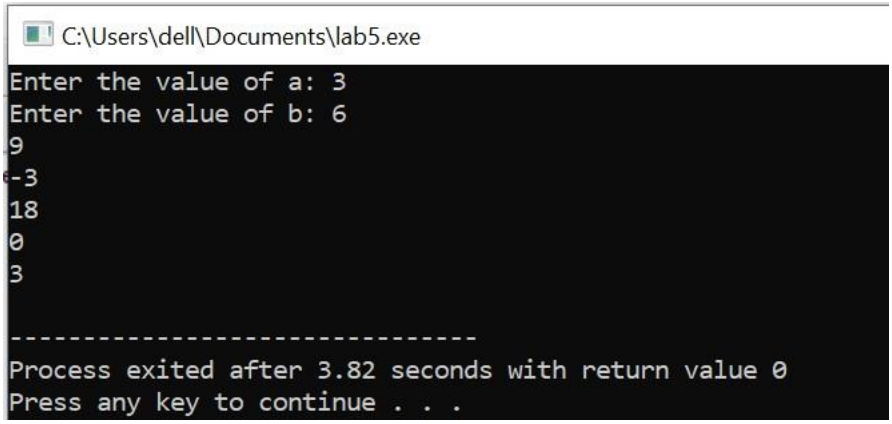
Take two integers as input from users apply arithmetic operation on them (+,-,*,/,%) as print them on screen.

```
#include<iostream>
using namespace std;
int main()
{
	int a,b,c,d,e,f,g;
cout<<"Enter the value of a: ";
cin>>a;
	cout<<"Enter the value of b:
";	cin>>b;	c=a+b;
cout<<c<<endl;  d=a-b;
```

```
        cout<<d<<endl;
e=a*b;
cout<<e<<endl;
f=a/b;
cout<<f<<endl;   g=a%b;
        cout<<g<<endl;
return 0;
}
```

Output:

```
Enter the value of a: 3
Enter the value of b: 6
9
-3
18
0
3
```



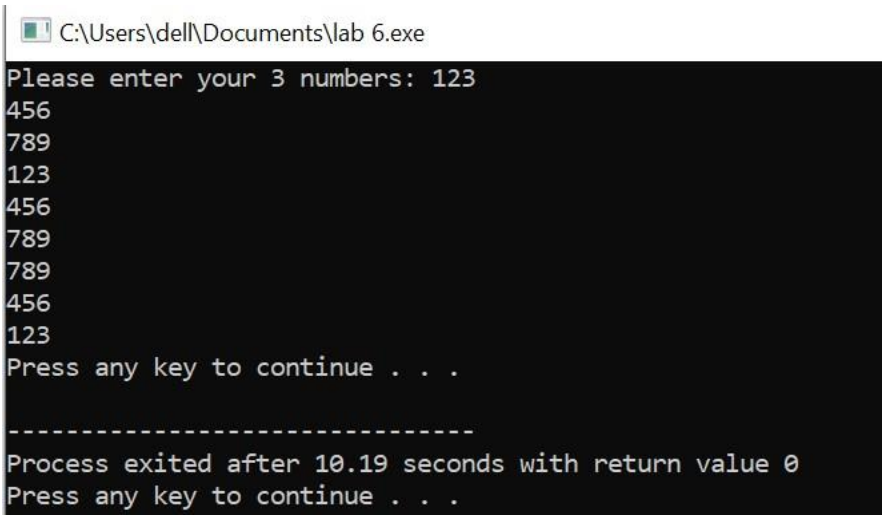
TASK3:

Write the output of the following code:

```
#include<iostream> using namespace std; int
main(){      int val1,val2,val3;       cout<<"Please
enter your 3 numbers: ";
cin>>val1>>val2>>val3;
cout<<val1<<"\n"<<val2<<"\n"<<val3<<"\n";
cout<<val3<<"\n"<<val2<<"\n"<<val1<<"\n";
system("PAUSE");       return 0;
}
```

Output:

```
Please enter your 3 numbers: 123
456
789
123
456
789
789
456
123
```



TASK4:

Write c++ program to calculate the distance between the two points.

NOTE:x1,y1,x2,y2 are all double values.

Formula:

$d=(x_2-x_1)^2+(y_2-y_1)^2$

```
#include<iostream>

using namespace std;

int main(){         int

x1,y1,x2,y2,d;

cout<<"Enter the
```

```
value of x1:";
cin>>x1;
        cout<<"enter the value of y1:";
cin>>y1;
        cout<<"enter the value of x2:";
cin>>x2;
        cout<<"enter the value of y2:";
cin>>y2;
        d=(x2-x1)*(x2-x1)+(y2-y1)*(y2-y1);
        cout<<"the distance between two point is: "<<d<<endl;
return 0;
}
```

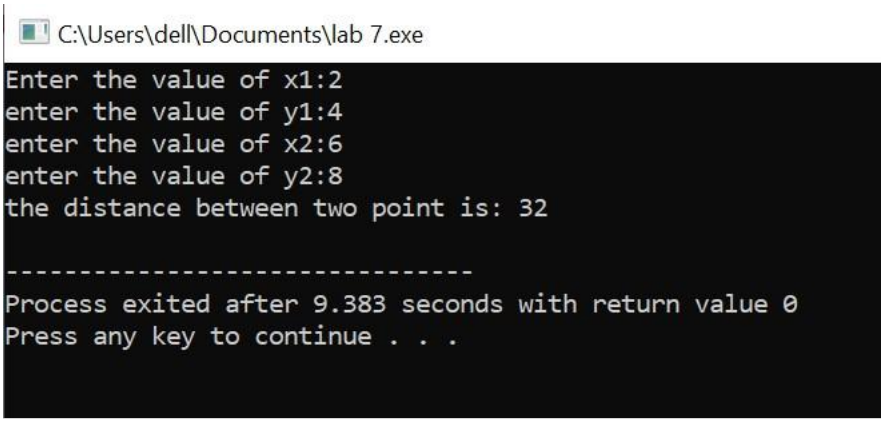
Output:

Enter the value of x1: 2 enter the

value of y1: 4 enter the value of x2: 6

enter the value of y2: 8

the distance between two point is: 32



TASK5:

1.calculate the area of circle(area=pi*r2)

2.calculate the area of a rectangle(area=length* width)

3.calculate the area of triangle(area=base*heig ht*.5)

```
#include<iostream>

using namespace std;

int main(){

//calculate the area of circle

float pi=3.141592;

float r,a;
```

```
cout<<"enter the value
of radius of a circle: ";

cin>>r;

a=pi*(r*r);

cout<<"Area of circle is:
"<<a<<endl;

//calculate the area of
rectangle

float b,l,w;

cout<<"enter the value
of length of rectangle: ";

cin>>l;

cout<<"enter the value
of width pf rectengel: ";

cin>>w;

b=l*w;

cout<<"the area of
rectangle is "<<a<<endl;

//calculate the area of a
triangle

float area,base,height;

cout<<"enter the value
of base of a triangle: ";

cin>>base;
```

```
cout<<"enter the value
of height of a triangle: ";

cin>>height;

area=base*height*.5;

cout<<"the area of
triangle is:
"<<area<<endl;

return 0;

}
```

Output:

```
Enter the value of radius
of a circle:3.4

Area of a circle
is:36.3168

Enter the value of
length of rectangle:6.5

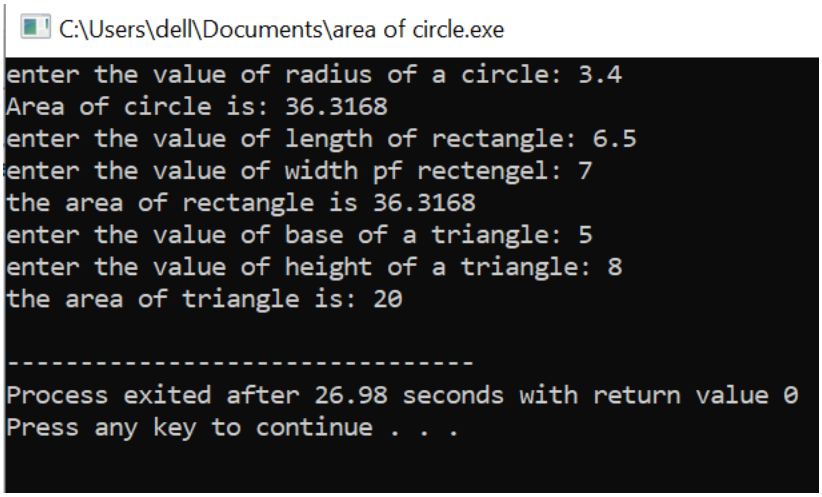
Enter the value of width
pf rectangle:7

The are of rectangle is
36.3168

Enter the value of base
of a triangle: 5

Enter the value of
height of a triangle: 8
```

The area of triangle is:
20



HOME TASK1:

Write a program that read in two variable and swaps the value of these variable.

```
#include<iostream>

using namespace std; int

main(){

        int var1,var2,tem;

cout<<"enter the value of var1: ";

cin>>var1;

        cout<<"enter the value of var2: ";

cin>>var2;

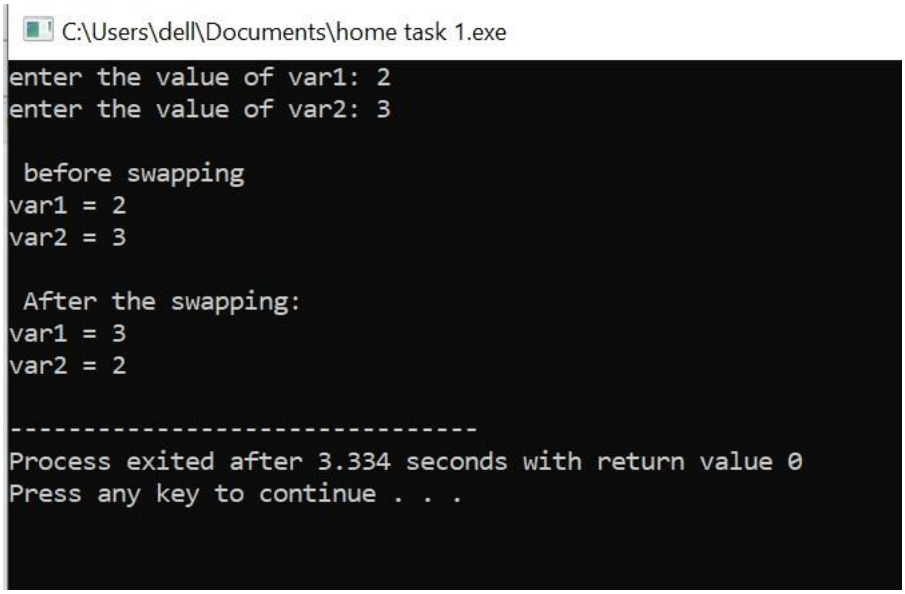
        cout<<"\n before

swapping"<<endl;          cout<<"var1 =
```

```
"<<var1<<endl;      cout<<"var2 =
"<<var2<<endl;      tem=var1;
var1=var2;   var2=tem;
        cout<<"\n After the swapping: "<<endl;
cout<<"var1 = "<<var1<<endl;   cout<<"var2 =
"<<var2<<endl;      return 0;
}
```

Output:

```
enter the value of var1: 2
enter the value of var2: 3
before swapping var1 = 2
var2 = 3 after swapping
var1 = 3 var2 = 2
```



HOME TASK2:

Make upper write a program that prompts the user to enter an alphabet in small caps (a,b,c,…,z) and display the entered alphabet into its upper caps (A,B,C,…,Z).

HINT:(ASCII)

```
#include<iostream>

using namespace std;

int main(){

        char letter;

        cout<<"enter a small letter:

";      cin>>letter;        letter=letter-

32;

        cout<<"uppercase letter= "<<letter<<endl;

return 0;


}
```

Output:

enter a small letter: a uppercase

letter= A

